

- 1

- i

- *

$$V_c(t) = V(1 - e^{-\frac{t}{RC}})$$

- *

$$V_c(RC) = V(1 - e^{-\frac{RC}{RC}})$$

- *

$$V_c(RC) = V(1 - e^{-1})$$

- *

$$V_c(RC) \approx 0.63V$$

- ii

- *

$$\left| \frac{V_c}{V_{in}} \right| = \frac{1}{\sqrt{1 + (2\pi fRC)^2}}$$

- * Using $F_c = \frac{1}{2\pi RC}$

- *

$$\left| \frac{V_c}{V_{in}} \right| = \frac{1}{\sqrt{1 + 1^2}}$$

- *

$$\left| \frac{V_c}{V_{in}} \right| = \frac{1}{\sqrt{2}}$$

- [[2]]

-

- By measuring using the ‘cursor’ tool on the oscilloscope, we can measured $\Delta t = 440\mu s$.

- Aka the time taken to reach 63% of the amplitude takes $440\mu s$.

- I will assume a $\pm 20\mu s$ uncertainty, this is due to the oscilloscope only measuring values in jumps of $40\mu s$. I choose half the differences between possible measurement values.

- According to part 1, the point in which 63% of the amplitude is reached is at $t = RC$

- We measured the resistance to be 4.6, with the multimeter. Below is the calculation without error propagation.

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$$RC = 4.6k\Omega \times 0.1F = 460\Omega f^{-1}$$

- Using python.our resistance measurement was taken with a multimeter which has a 1.2% uncertainty. The capacitor specifies only its tolerance and does not have an uncertainty, so a 1% uncertainty has been assumed.

- `R = ufloat(4600, 4600 * 0.012)`

- `C = ufloat(0.1e-6, 0.1e-6 * 0.01)`

- `Tcalculated = R * C`

- `print(Tcalculated)`

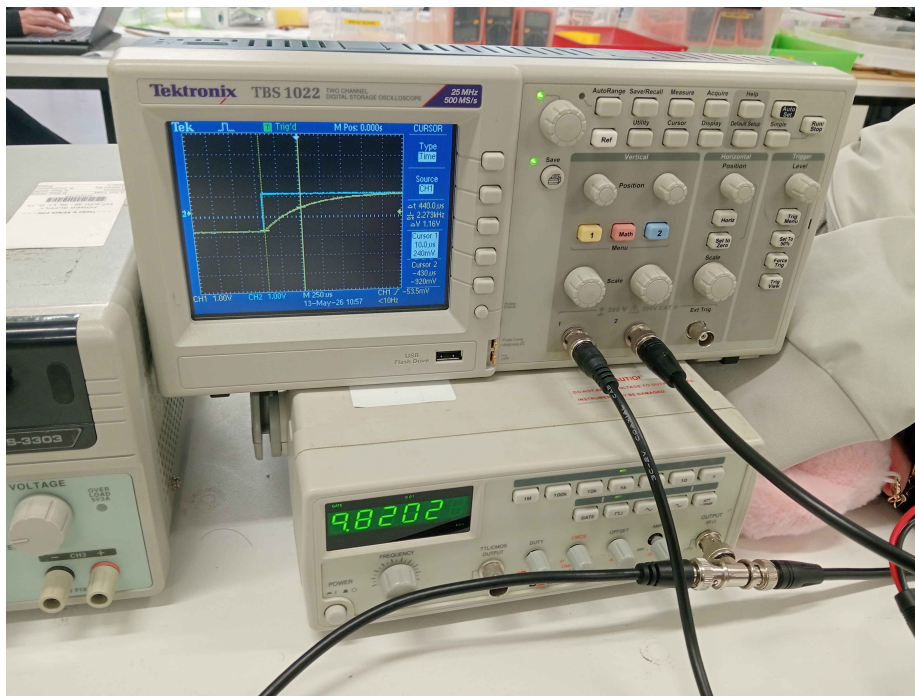


Figure 1: Photo of our oscilloscope for part 2

- The return value is $(4.60 \pm 0.07) \times 10^{-4}$. *note: The uncertainties library has been imported and is used for all further python calculations
- Now we have both the measured and calculated values of the milliseconds, we can extrapolate a σ uncertainty for both the values.

$$\sigma = \frac{|t_{\text{measured}} - t_{\text{calculated}}|}{\sqrt{\sigma_1^2 + \sigma_2^2}}$$

- again, using python, and our measured 440 nanoseconds, we can calculate the sigma.
- `TMeasured = ufloat(440e-9, 20e-9)`
`Sigma = (TCalculated.n - TMeasured.n) / (TCalculated + TMeasured).s`
`print(Sigma)`
- the return value for this is 0.9411057177286997
- Since this is < 3 , we can safely say that our two values corroborate each other.

• 3

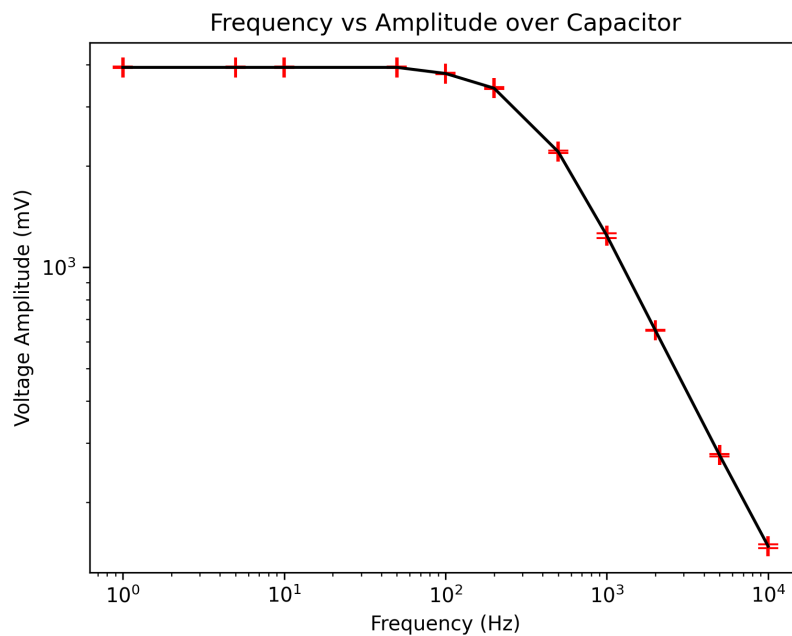


Figure 2: Frequency vs Amplitude over Capacitor

- Frequency uncertainty is 1%, and voltage amplitude uncertainty is half of the lowest discrete jump. So if the oscilloscope measures 1140mV, the jump between each value is 20mV.

- I'm going to interpolate and say the cutoff frequency is a point on 10^2 Hz. I will assume a

$$\sqrt{10} \times 10^2 \text{ Hz}$$

- uncertainty, just due to how much uncertainty our log graph gives us.
- Calculating F_c in python using $F_c = \frac{1}{2\pi RC}$:

```
print(1/(2 * math.pi * R * C))
```
- we get $(3.46 \pm 0.05) \times 10^2$
- This value, aka 346Hz, is well within our measured $10^2 \pm \sqrt{10} \times 10^2 \text{ Hz}$
- using python

```
FCalculated = 1/(2 * math.pi * R * C)
FMeasured = ufloat(1e2, (10 ** 0.5) * 1e2)
print(FMeasured)
Sigma = (FCalculated.n - FMeasured.n) / (FCalculated + FMeasured).s
print(Sigma)
```
- We get a measured sigma of 0.7777
- Since our sigma is < 3 , we can say the values corroborate each other.

• 4

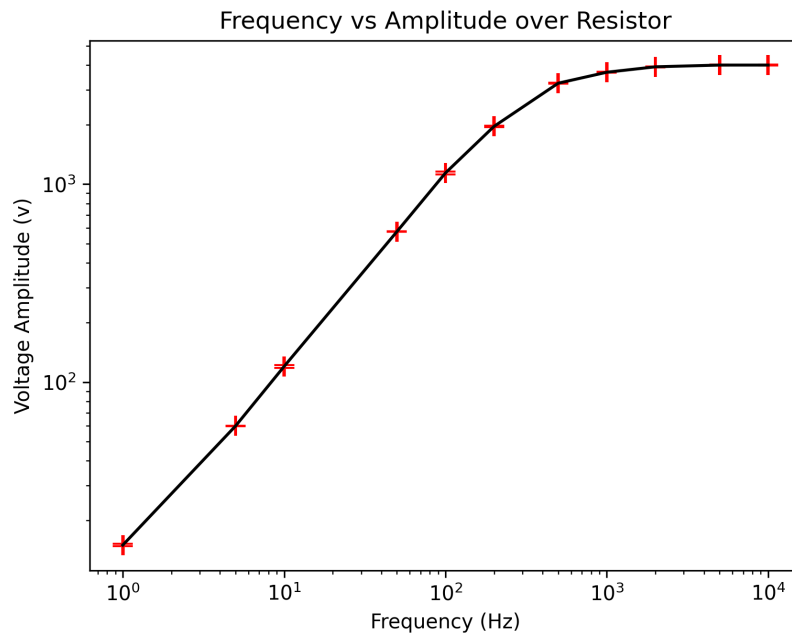


Figure 3: Frequency vs Amplitude over Resistor

- Similarly to part 3, Frequency uncertainty is 1%, and voltage amplitude uncertainty is half of the lowest discrete jump. So if the oscilloscope measures 1140mV, the jump between each value is 20mV.
- this corroborates with our first graph. At higher frequencies, the capacitor has a voltage that converges to 0, which means a voltage amplitude of V, aka voltage in. and at lower frequencies the capacitor has a voltage amplitude converging to 0, since the capacitor will just have the voltage of the V_{in} .
- we can also visually see that the cutoff frequency is when the voltage sort of hits its ceiling.
- The name I can [[think]] of the circuit is just the name of the graph: ‘Frequency vs Amplitude over Resistor’.
- If i were to think of a name that sort of describes the information/meaning of the results it would be something more on the lines of ‘How frequency will cause the voltage of a capacitor to approach 0’.